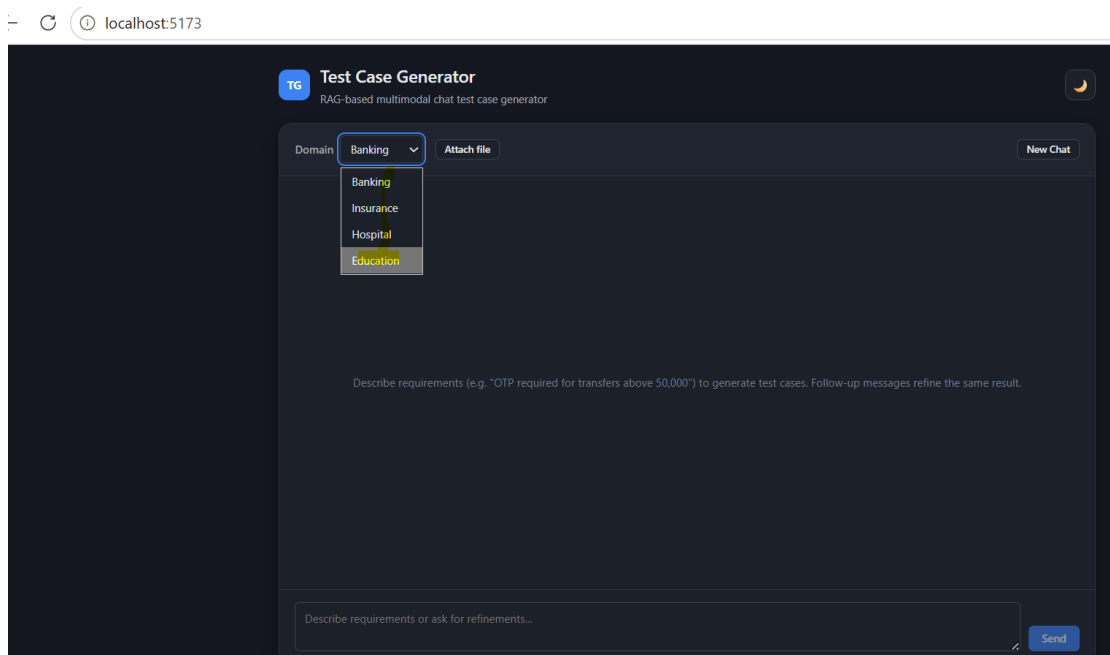


Assignment

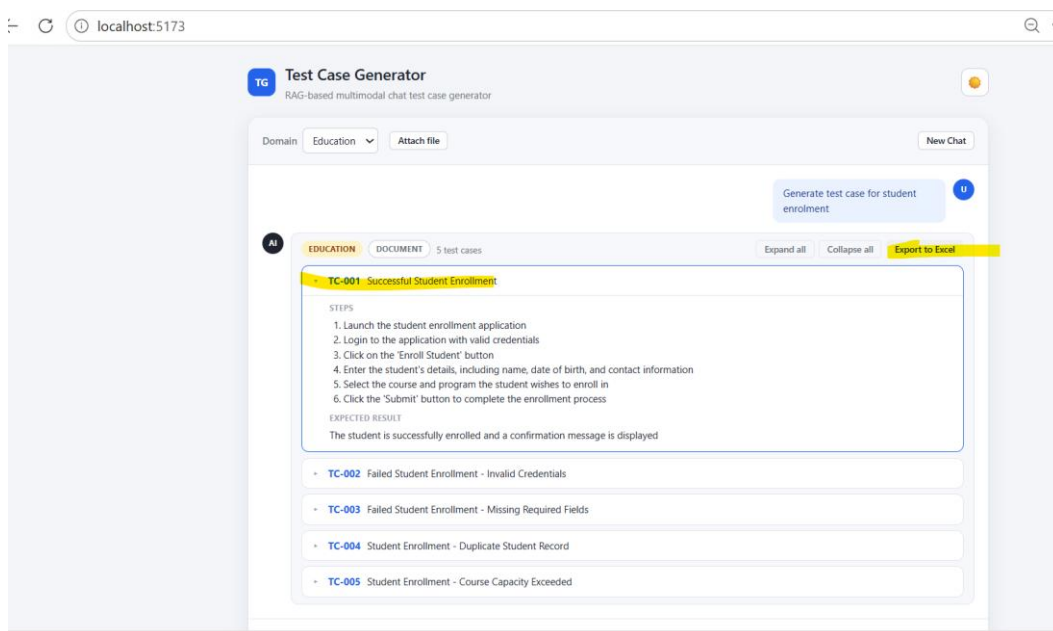
Phase-2, Week-5, Day1 and Day2

- 1) Improve the application with a modern, responsive UI, Light/Dark mode toggle.
- 2) Expand support Education domains with Banking, Insurance, Hospital
- 3) Enable users to download generated test cases in a structured Excel (.xlsx) format.

New Education domain



Test cases



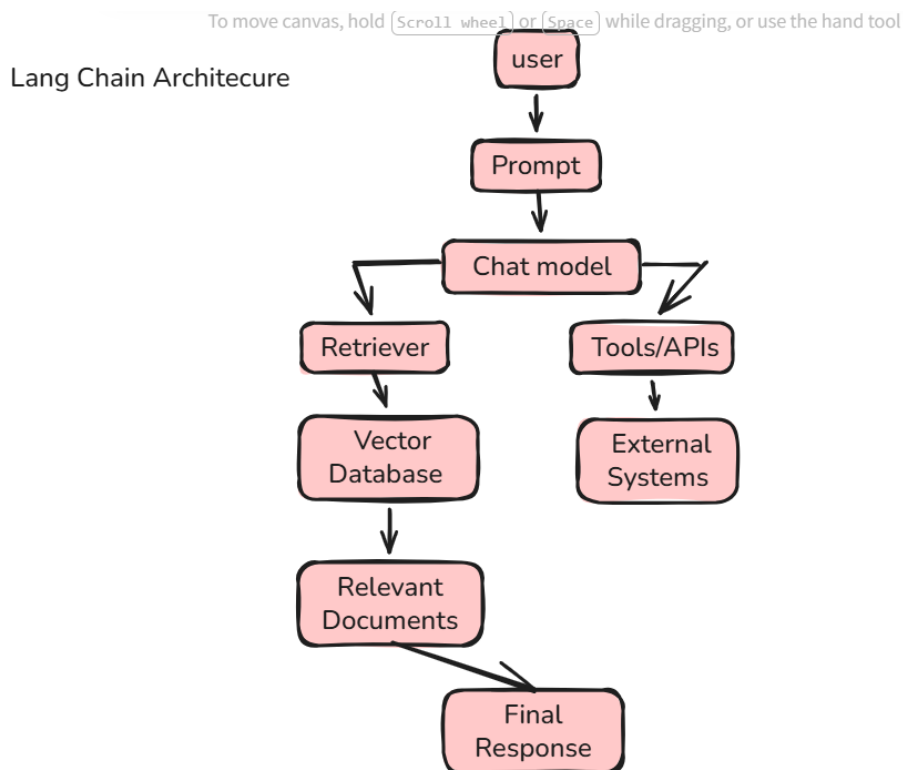
Export to Excel

The screenshot shows an Excel spreadsheet with the following data:

ID	Title	Steps	Expected Result
TC-001	Successful Student Enrollment	1. Launch the student enrollment application 2. Login to the application with valid credentials 3. Click on the 'Enroll Student' button 4. Enter the student's details, including name, date of birth, and contact information 5. Select the course and program the student wishes to enroll in 6. Click the 'Submit' button to complete the enrollment process	The student is successfully enrolled and a confirmation message is displayed
TC-002	Failed Student Enrollment - Invalid Credentials	1. Launch the student enrollment application 2. Login to the application with invalid credentials 3. Click on the 'Enroll Student' button 4. Enter the student's details, including name, date of birth, and contact information 5. Select the course and program the student wishes to enroll in 6. Click the 'Submit' button to complete the enrollment process	An error message is displayed indicating that the login credentials are invalid
TC-003	Failed Student Enrollment - Invalid Credentials	1. Launch the student enrollment application 2. Login to the application with valid credentials 3. Click on the 'Enroll Student' button 4. Enter the student's details, leaving out required fields such as name	An error message is displayed indicating that required fields are missing

Lang chain Explore

Lang Chain Architecture



1. Models (The Brain)

What is it?

A **Model** is the AI that generates responses.

Think of it as the **brain** of your application.

Examples:

- OpenAI GPT
- Mistral
- Claude
- Gemini
- Llama

Example:

User: Generate test cases for login page. -> GPT-5 / Mistral -> Returns test cases.

2. Prompts (Instructions to the AI)

A prompt tells the model **how to behave**.

Instead of writing a long string every time, LangChain lets you create reusable templates.

3. Documents

Documents represent pieces of information.

Documents contain:

- page_content
- metadata

They are the basic unit used in retrieval systems.

4. Document Loaders

A loader reads files.

Supported examples include:

- PDF
- DOCX
- CSV
- Excel
- Web pages

5. Text Splitters

LLMs cannot process huge documents efficiently.

A splitter breaks a large document into smaller chunks.

6. Embeddings

Embeddings convert text into numbers (vectors).

Embedding models include:

- Mistral Embed
- OpenAI Embeddings
- Gemini Embeddings

7. Vector Store

A Vector Store stores embeddings.

Examples:

- MongoDB Atlas Vector Search
- Pinecone
- Chroma
- FAISS

Instead of storing only text

8. Retriever

The Retriever searches the vector database.

The retriever does **not** generate an answer.

It only finds relevant documents.

9. RAG (Retrieval-Augmented Generation)

This is the pattern you're building.

User Question

↓

Embedding

↓

Vector Search

↓

Relevant Documents

↓

LLM

↓

Answer

10. Chains

A Chain connects multiple components.

Instead of writing lots of code manually, LangChain lets you compose these steps into a pipeline. LangChain's core abstractions are designed to be modular so components can be combined easily.

11. Memory

Memory stores conversation history.

12. Tools

Tools let the AI perform actions.

Examples:

- Weather API
- Calculator
- Database
- Google Search
- REST APIs

13. Agents

An Agent decides **what to do next**.

Instead of following a fixed pipeline, it reasons and chooses the appropriate tool.

Agents are useful when the workflow isn't predetermined. LangChain provides a high-level agent interface built on LangGraph for orchestration.

14. Output Parser

Models usually return text.

An Output Parser converts it into structured data.

Useful when you need:

- JSON
- CSV
- XML
- Structured test cases

Complete Flow (Your Project)

Your AI Test Case Generator can follow this architecture:

User Requirement

|

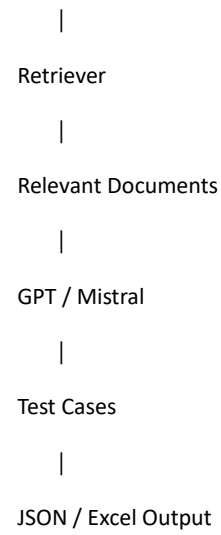
Prompt Template

|

Embedding Model (Mistral Embed)

|

MongoDB Atlas Vector Search



This is a classic RAG (Retrieval-Augmented Generation) workflow and is one of the most common patterns described in the LangChain component architecture.